

Source Recursive Expected Utility and Process Preference on source mixture Sets

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Abstract

Decision making often involves the compounding of risks from different sources. We introduce a choice domain called *source mixture set* which associates sources of uncertainty to different mixture operations, and characterize *source recursive expected utility* (SREU) theory on this domain by imposing a new axiom of Within Source Independence together with completeness, transitivity, and continuity. Under SREU, when a decision maker consistently prefers risks from one source over those from another, she prefers to resolve risks more decisively using the preferred source. This prediction is supported by the results of a laboratory experiment. In an application to portfolio choice, we demonstrate that the SREU investor exhibits home bias if she consistently prefers home risks over identically distributed foreign risks.

Keywords: mixture sets, source preference, process preference, reduction of compound lottery, home bias

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1 Introduction

Decision making often involves the compounding of risks from multiple sources. When introducing new policies, governments consider the reactions of different stakeholders, e.g., the public, experts, and foreign governments. Likewise, an investor crafting her portfolio must contend with risks inherent in its various segments, such as domestic and foreign investments.

The idea that risky prospects (or simply risks), identical in terms of their distribution of outcomes but arising from different sources, may be valued differently first appears in a question posed in Keynes (1921), *“If two probabilities are equal in degree, ought we, in choosing our course of action, to prefer that one which is based on a greater body of knowledge?”*¹ Such behavior, referred to as source preference, has been tested in a growing experimental literature since the early works of Heath and Tversky (1991) and Fox and Tversky (1995).² In this regard, Heath and Tversky further suggest that investors may sometimes be *“willing to forego the advantage of diversification and concentrate on a small number of companies with which they are presumably familiar”*.

With the prevalence of compound risks from multiple sources and ample

¹This quote summarizes what Keynes wrote earlier in his treatise, “... in the first case we know that the urn contains black and white in equal proportions; in the second case the proportion of each colour is unknown, and each ball is as likely to be black as white. It is evident that in either case the probability of drawing a white ball is $1/2$,...”; this unknown urn idea reappears in Ellsberg’s (1961) two-urn thought experiment.

²These include Chew et al. (2008), Abdellaoui et al. (2011), Armantier and Treich (2016), and Chew et al. (2023), among others.

empirical support for source-sensitive behavior, it would be valuable to understand this type of decision-making with a formal model. To this end, we introduce a new choice domain, namely *source mixture set*, and axiomatize on this domain a preference representation that we call *source recursive expected utility*. The key departure from a standard model is that we allow the *reduction of compound lottery axiom* (RCLA) to fail for compound risks from multiple sources,³ though we assume it still holds within a single source of risk, i.e., the decision maker is indifferent between compound lotteries arising from the same source and their actuarial reductions.

An intuitive way to understand the failure of RCLA is that the process of resolution may lead the decision maker to treat actuarially equivalent risks differently depending on how they are resolved. This modeling assumption is supported by evidence in Chew et al. (2023) which finds tight links between each pair of ambiguity, compound risk, and source-dependent natural-event risk attitudes.⁴ Furthermore, two-thirds of more than 3000 subjects exhibit a non-neutral attitude towards all three types of uncertainty relative to their

³Following Halevy (2007), evidence of non-RCLA behavior has been reported in Abdellaoui et al. (2015), Chew et al. (2017), Dean and Ortoleva (2019), and Gillen et al. (2019), among others.

⁴Ellsberg’s (1961) thought experiment involving bets on urns with known and unknown compositions explicates ambiguity aversion, offering the distinction between risk and ambiguity. This phenomenon may be explained by source preference in another way: Bets on each of the two urns are treated as 50-50 lotteries, with ambiguity aversion arising from a preference for lotteries determined by a source of risk based on more knowledge (e.g., Keynes, 1921; Smith, 1969; Chew and Sagi, 2008).

benchmark risk attitudes. Such choice patterns support our intuitive approach to linking the failure of RCLA to source preference.

Outline of the Model and Applications. The construction of our new domain is based on Herstein and Milnor (1953), who offer an elegant axiomatization of expected utility (EU) theory on *mixture sets* traceable to von Neumann and Morgenstern (1944). A mixture set $\mathcal{M}$ is endowed with a single mixture operation $\alpha a \oplus (1 - \alpha)b$ with a and b being members of $\mathcal{M}$ and α a real number in $[0, 1]$. When elements of $\mathcal{M}$ are lotteries, and the mixture is probabilistic, the operation results in a lottery equivalent to the compound lottery delivering a with probability α and b with probability $1 - \alpha$.

To incorporate source preference, we use a collection of mixture operations to define a source mixture set, whose elements are referred to as *source lotteries*, with each mixture operation modeling (and indexed by) a source of risk s in a family of sources S . For example, we can construct a source mixture set $\mathcal{M}$ with two sources of risk and a space of sure outcomes such that $\mathcal{M}$ captures a rich variety of risky prospects beyond the usual setting of two-stage compound lotteries (Example 2, Section 2.1).

As illustrated in Figure 1, an element $c = \alpha a \oplus_s (1 - \alpha)b$ of a source mixture set $\mathcal{M}$ can be interpreted as a risky prospect—delivering a with probability α and otherwise delivering b —in which the risk arises from source s in S . The two outcomes a and b are themselves source lotteries as they are elements of $\mathcal{M}$ and therefore c contains further information about how the risk resolves within a and b . For example, in Figure 1, they are both risky prospects with two possible outcomes from the sources of risk u and v , respectively.

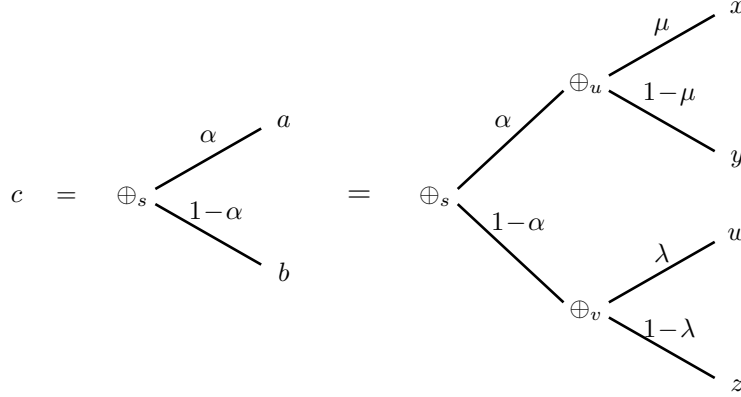


Figure 1: An example of source lottery

Abstracting away the state space, pure consequences, and the number of stages, source mixture sets and source lotteries provide a novel, parsimonious, and flexible structure to model the compounding of risks from multiple sources.

For preferences on source mixture sets, we retain the first two axioms of ordering and continuity of Herstein and Milnor (1953) and relax their axiom of independence by only requiring it to hold for risks from the same source. We call this *Within Source Independence*. This yields an axiomatization of a *source recursive expected utility* (SREU) model, which can accommodate non-RCLA behavior arising from source preference.

In relaxing RCLA, SREU can exhibit a wide range of process preferences—*how* compound risks resolve matters when they are actuarially equivalent—including being non-neutral towards the order of sources in otherwise identical source lotteries. Say that a source s is preferred to another t when there is a consistent preference for the same risks arising from the source s . We characterize in Proposition 1 a *preference for decisiveness of preferred source* (PDPS) in preferring processes that resolve “more” risk with the preferred

source. Our definition of PDPS involves a comparison between two source lotteries in which the order of two sources of risk is switched, which means that such a definition cannot be accommodated by conventional two-stage compound lotteries. We provide evidence to support PDPS with a simple laboratory experiment in Section 5.1.

Relation to the Literature. Source mixture sets are closely related to other choice domains in the literature. In intertemporal decision making, Kreps and Porteus (1978) offer temporal lotteries in which the uncertainty is “dated” by the timing of its resolution. This enables their modeling of non-neutral attitude towards the timing of uncertainty resolution as part of their development of a recursive preference model without imposing an analogous RCLA in the temporal setting. Segal (1990) relaxes RCLA in the domain of atemporal compound lotteries. In allowing equally distributed risks arising from different stages to be valued differently, RCLA may not hold under the assumptions of within-stage independence and recursive evaluation of compound lotteries. Dillenberger (2010) captures a preference for immediate resolution of uncertainty based on recursive evaluation of compound lotteries together with a stage-invariant non-EU risk preference. While source preference is not explicitly mentioned on these domains, these papers discuss behavioral patterns that relate to those in the present paper.

Most theories of source preference do not consider source-based process preference as is done in this paper. These include Chew and Sagi (2008), Gul and Pesendorfer (2015), and Cappelli et al. (2020). Both Chew and Sagi (2008) and Gul and Pesendorfer (2015) take, as primitive, a preference over Savagean

acts and identify sources of risk with collections of events. Cappelli et al. (2020) share with us the assumption of an exogenous set of sources. Their key premise is that the same deterministic outcome arising from different sources may be valued differently, captured by their notion of “subjective prices”.

Our paper is also related to a strand of literature that explains ambiguity aversion by using a second-order belief over priors and relaxing RCLA for the resulting compound lotteries (e.g., Becker and Brownson, 1964; Segal, 1987; Klibanoff et al., 2005; Seo, 2009). With a built-in number of stages and thus a fixed ordering of sources of risk, the setting of multi-stage compound lotteries (including *subjective* compound lotteries residing on a product state space, e.g., Nau, 2006; Ergin and Gul, 2009) is not sufficiently flexible for the analysis of source-sensitive decision making under compound risks.⁵ This explanation of ambiguity aversion can be accommodated by our model as second-order beliefs and priors could be considered as different sources of risk.

For the remainder of the paper, Section 2 provides the formal definition of a source mixture set together with two examples. Axioms and the representation theorem are developed in Section 3. Using SREU, we model process preference in Section 4. Section 5 comprises our experimental test of PDPS and an account of home bias in international finance. Section 6 concludes. Proofs are relegated to an appendix.

⁵Epstein and Halevy (2019) goes beyond two sources and shows that an additional source of uncertainty about how color composition in two unknown urns are correlated leads to substantial ambiguity-averse behavior when choosing between bets on a single unknown urn versus bets on both unknown urns.

2 Source Mixture Set

As discussed in the Introduction, we offer a formal definition below of the a *source mixture set* $\mathcal{M}$, involving a family S of mixture operations.

Definition 1. We say $(\mathcal{M}, \{\oplus_s\}_{s \in S})$ is a **source mixture set** if each mixture operation $\mu a \oplus_s (1 - \mu)b$ satisfies:

1. $1a \oplus_s (1 - 1)b = a$,
2. $\mu a \oplus_s (1 - \mu)b = (1 - \mu)b \oplus_s \mu a$,
3. $\lambda[\mu a \oplus_s (1 - \mu)b] \oplus_s (1 - \lambda)b = \lambda\mu a \oplus_s (1 - \lambda\mu)b$,

for any $a, b \in \mathcal{M}$ and any $\mu, \lambda \in [0, 1]$.

An element $c = \alpha a \oplus_s (1 - \alpha)b$ of a source mixture set $\mathcal{M}$ is called a source lottery with two outcomes a and b , which are themselves source lotteries. Consequently, c contains further information about how risk resolves within a and b . For example, in Figure 1, they are both risky prospects from two different sources of risk, u and v , with each delivering two possible outcomes.

To understand how source mixture sets generalize the standard mixture sets, compare Condition 3 of Definition 1 with the *mixture reduction* property below:

$$\lambda[\mu a \oplus_s (1 - \mu)b] \oplus_t (1 - \lambda)b = \lambda\mu a \oplus_t (1 - \lambda\mu)b$$

for any $a, b \in \mathcal{M}$, any $\mu, \lambda \in [0, 1]$, and any $s, t \in S$. We can see that in our definition of source mixture sets, mixture reduction holds only for risks from the same source. Assuming it holds regardless of the underlying sources of risk

as in the above condition returns us to a standard mixture set, i.e., $\oplus_s = \oplus_t$ for any $s, t \in S$.

2.1 Examples of source mixture Sets

Anscombe and Aumann (1963), and subsequently Seo (2009), adopt a choice domain $\mathcal{L}^2$ consisting of lotteries over acts, referred to as horse lotteries, where acts are functions from a finite state space S to a set $\mathcal{L}^1$ of lotteries over a set X of sure prizes, which are referred to as roulette lotteries. In his characterization of preferences involving second-order beliefs, Seo (2009) considers two mixture operations on $\mathcal{L}^2$ where the first-stage mixture operation is standard, while the domain of the second-stage mixture operation is restricted to degenerate horse lotteries, and thus is mixing two acts state-by-state. In the example below, we extend this restricted second-stage mixture operation to the whole $\mathcal{L}^2$, enabling us to view it as a source mixture set.

Example 1. *For simplicity, we assume that $\mathcal{L}^1$ and $\mathcal{L}^2$ both consist of simple lotteries, i.e., probability distributions with finite support. Elements of $\mathcal{L}^2$ are denoted P and Q . The first-stage mixture operation $\alpha P \oplus_1 (1 - \alpha)Q$ on $\mathcal{L}^2$ is defined as usual, i.e., the probability of any act f being delivered by $\alpha P \oplus_1 (1 - \alpha)Q$ is equal to $\alpha P(f) + (1 - \alpha)Q(f)$. The second-stage mixture operation $\alpha P \oplus_2 (1 - \alpha)Q$ is defined as follows: for any act f in the support of P and any act g in the support of Q , the probability of the act $\alpha f \oplus (1 - \alpha)g$, defined by state-by-state mixture between lotteries of $\mathcal{L}^1$, being realized is $P(f) \cdot Q(g)$.⁶*

⁶This definition is in the spirit of mixture operations commonly adopted in the

In the next example, we construct a source mixture set $\mathcal{M}$ with two sources of risk $S = \{s_1, s_2\}$ and a space of sure outcomes X . Notably, the constructed $\mathcal{M}$ contains a rich assortment of risky prospects beyond the space of two-stage compound lotteries, typically adopted to study non-RCLA behavior (e.g., Segal, 1990; Dillenberger, 2010).

Example 2. Let $\mathcal{L}_s^0 = \{(x, s) : x \in X\}$ for each of the two sources $s \in S = \{s_1, s_2\}$.⁷ Iteratively for each integer $k > 0$ and for each source $s \in S$, let $\mathcal{L}_s^k$ be the space of non-degenerate simple lotteries over $\bigcup_{0 \leq l \leq k-1} \mathcal{L}_t^l$ from source s (with $t \neq s$). Notice that under this formulation the prizes of a lottery from source s are lotteries or sure outcomes from the other source t . Then, we can define the mixture operation $\alpha P \oplus_s (1 - \alpha)Q$ on $\mathcal{L}_s = \bigcup_{k \geq 1} \mathcal{L}_s^k \cup \mathcal{L}_t^0$ (for each $s \in S$ with $t \neq s$) in the usual manner. Finally, $\mathcal{M} = \mathcal{L}_{s_1} \cup \mathcal{L}_{s_2}$ is a source mixture set with the two mixture operations $\oplus_{s_1}$ and $\oplus_{s_2}$ under a natural extension. That is, in performing $\alpha P \oplus_s (1 - \alpha)Q$ where $P \in \mathcal{L}_t$ (and/or $Q \in \mathcal{L}_t$), treat P (and/or Q) as a (though non-existing) degenerate lottery in $\mathcal{L}_s$.

This example illustrates how a source mixture set enables modeling of a broader range of behavior over compound risks than is allowed by conven-

literature on menu preference since Dekel et al. (2001) where the decision maker can choose from combinations drawn independently from each of two menus when mixing them.

⁷The second dimension of the elements in $\mathcal{L}_s^0$ is introduced to define the following $\mathcal{L}_{s_1}$ and $\mathcal{L}_{s_2}$ to be distinct sets. Our theory does not rely on individuals showing a strict preference between some (x, s_1) and (x, s_2) to distinguish between these two sources of risk.

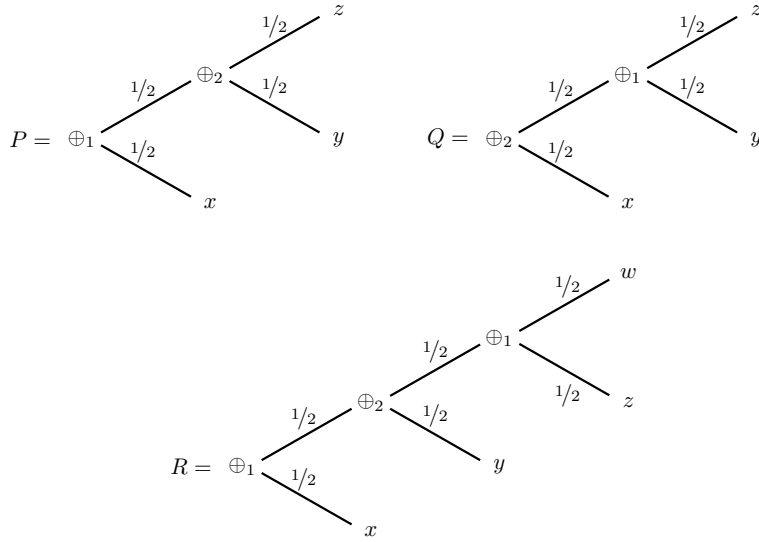


Figure 2: source lotteries in Example 2

tional compound lotteries. For example, source lotteries P and Q in Figure 2 cannot be simultaneously accommodated in a setting of two-stage compound lotteries, as they differ only in the order of the two sources of risk. In other words, P and Q each belong to their own distinct two-stage compound lotteries domain. Beyond two-stage compound lotteries, consider source lottery R with a repeated instance of source 1 and a single instance of source 2. In the context of publishing, source 1 could represent an editor, with source 2 representing referees. This source lottery R cannot be accommodated even in the space of three-stage compound lotteries, since the editor, as a source of risk, appears twice in its resolution process. This is true in general for any domain of multi-stage compound lotteries with repeated sources of risk, for example in modeling more complex scenarios such as revise-and-resubmissions, which would involve further instances of source 2 (same referees) and source 1 (same editor).

Source lottery R in Figure 2 with repeated source 1 can represent editor, with source 2 being referees. Editor first decides to send to referees or not, referees make decisions/reports, then same editor looks at reports for final decision.

3 Source Recursive Expected Utility

The following axioms on a preference ordering $\succsim$ on a source mixture set $\mathcal{M}$ are based on those in Herstein and Milnor (1953).

Axiom 1. (*Ordering*) *The preference ordering $\succsim$ is complete and transitive.*⁸

Axiom 2. (*Continuity*) *For any $a, b, c \in \mathcal{M}$ and for any $s \in S$, $\{\alpha : \alpha a \oplus_s (1 - \alpha)b \succsim c\}$ and $\{\alpha : c \succsim \alpha a \oplus_s (1 - \alpha)b\}$ are closed.*

The following Within Source Independence is a relaxation of Herstein and Milnor's (1953) Axiom 3.

Axiom 3. (*Within Source Independence*) *For any $a, b, c \in \mathcal{M}$ and for any $s \in S$, $a \sim b$ implies $\frac{1}{2}a \oplus_s \frac{1}{2}c \sim \frac{1}{2}b \oplus_s \frac{1}{2}c$.*

To facilitate comparison with EU, observe that the standard axiom of Independence may be stated in a source mixture set as follows: for any $a, b, c \in \mathcal{M}$ and for any $s, t \in S$, $a \sim b$ implies $\frac{1}{2}a \oplus_s \frac{1}{2}c \sim \frac{1}{2}b \oplus_t \frac{1}{2}c$. This is weakened by Within Source Independence by requiring s and t to be the same source.

Now, we define an SREU representation.

⁸We say $\succsim$ is complete if $a \succsim b$ or $b \succsim a$ for all $a, b \in \mathcal{M}$. We say $\succsim$ is transitive if $a \succsim b$ and $b \succsim c$ implies $a \succsim c$ for all $a, b, c \in \mathcal{M}$.

Definition 2. We say a preference ordering $\succsim$ on a source mixture set $\mathcal{M}$ has an SREU representation if there exists a utility function $U : \mathcal{M} \rightarrow \mathbb{R}$ such that $U(a) \geq U(b)$ if and only if $a \succsim b$ for each $a, b \in \mathcal{M}$, and a family $\{T_s : U(\mathcal{M}) \rightarrow \mathbb{R}\}_{s \in S}$ of strictly increasing continuous transformation functions with a common intersection point, i.e., there exists $u \in U(\mathcal{M})$ with $T_s(u) = T_t(u)$ for all $s, t \in S$, such that

$$U(\alpha a \oplus_s (1 - \alpha)b) = T_s^{-1}(\alpha T_s \circ U(a) + (1 - \alpha)T_s \circ U(b))$$

for all $a, b \in \mathcal{M}$ and all $0 \leq \alpha \leq 1$.

We illustrate the recursive nature of SREU using $c = \alpha a \oplus_s (1 - \alpha)b$ with $a = \mu x \oplus_u (1 - \mu)y$ and $b = \lambda w \oplus_v (1 - \lambda)z$ mentioned in Figure 1 earlier. Given utilities $U(x), U(y), U(w)$, and $U(z)$, the utility of c can be arrived at recursively by deriving $U(a)$ and $U(b)$ followed by $U(c)$:

$$\begin{aligned} U(a) &= T_u^{-1}(\mu T_u \circ U(x) + (1 - \mu)T_u \circ U(y)), \\ U(b) &= T_v^{-1}(\lambda T_v \circ U(w) + (1 - \lambda)T_v \circ U(z)), \\ U(c) &= T_s^{-1}(\alpha T_s \circ U(a) + (1 - \alpha)T_s \circ U(b)). \end{aligned}$$

It is worth pointing out that the recursivity of SREU comes from the Within Source Independence axiom as well as the structure of the source mixture set, in which the domain of each mixture operation is the whole source mixture set $\mathcal{M}$.

Regarding the examples in Section 2.1, we observe that Seo's (2009) second-

order subjective expected utility representation is not an SREU representation. This is because Seo's second-stage independence does not hold on the whole choice domain, but is rather required to hold only for roulette lotteries (degenerate horse lotteries which are also constant acts). This is in contrast to our Within Source Independence axiom which holds on the entire source mixture set.

An SREU representation for Example 2 (Section 2.1). *Suppose u_1 and u_2 are two continuous and strictly increasing functions from $X = [w, b]$ to $\mathbb{R}$ with an intersection. For the certain outcomes in $\mathcal{L}_1^0 \cup \mathcal{L}_2^0$, let $c((x, s)) = x$ for each $x \in X$ and each $s \in S = \{s_1, s_2\}$. For any $s \in S$ and iteratively for each $k > 0$, let certainty equivalence of a source lottery $P \in \mathcal{L}_s^k$ be given by*

$$c(P) = u_s^{-1} \left(\sum_{Q \in \text{supp}(P)} p(Q) u_s \circ c(Q) \right).$$

Then, c is a well-defined certainty equivalence function on the whole source mixture set $\mathcal{M}$. This preference ordering represented by c has an SREU representation. Simply, let $U = c$ and $T_s = u_s$ for each $s \in S$.

We now present our representation theorem.

Theorem 1. *A preference ordering $\succsim$ on a source mixture set $\mathcal{M}$ satisfies axioms 1-3 if and only if it has an SREU representation.*

The theorem is proved in a constructive manner. In particular, the transformation functions are constructed using independently derived utility functions for each source of risk.

Compared to the EU representation by a single utility function U , the SREU representation of a preference relation $\succsim$ is augmented by a collection of transformation functions $\{T_s\}_{s \in S}$ intersecting with each other at some value u in $U(\mathcal{M})$. It is instructive to discuss the uniqueness property of SREU representations of $\succsim$ compared to the uniqueness class of EU representations defined by positive affine transformations.

Fixing the utility function U , it is clear that each transformation function T_s can be affine transformed into $\alpha T_s + (1 - \alpha)T_s(u)$ ($\alpha > 0$) such that the collection of transformation functions still intersects at u . Consider another utility function $\tilde{U} = T \circ U$ that also represents the preference relation $\succsim$ where T is a strictly increasing continuous function. It follows that $(\tilde{U}, \{T_s \circ T^{-1}\}_{s \in S})$ is also an SREU representation of $\succsim$ in which $\{T_s \circ T^{-1}\}_{s \in S}$ intersect with each other at $T(u)$. It follows that the composite function $T_s \circ U$ captures the decision maker's risk attitude for the source of risk s .

4 Process Preference

When the source lotteries from one source s are consistently preferred to identically distributed source lotteries from another source t (e.g., s may be more familiar than t to the decision maker), we say that the source s is preferred to the source t . We formally state this definition of a partial ordering over sources of risk below.

Definition 3. *For any two sources $s, t \in S$, we say the source s is weakly*

preferred to the source t if it is the case that

$$\alpha a \oplus_s (1 - \alpha)b \succsim \alpha a \oplus_t (1 - \alpha)b$$

for any $a, b \in \mathcal{M}$ and any $\alpha \in [0, 1]$.

We can also define a strict preference for one source s to another t by requiring a strict preference $\alpha a \oplus_s (1 - \alpha)b \succ \alpha a \oplus_t (1 - \alpha)b$ in the range of $\alpha \in (0, 1)$ for any two source lotteries $a, b \in \mathcal{M}$ between which the decision maker is not indifferent.

Maintaining the axioms of SREU, we could equivalently restrict $\alpha = \frac{1}{2}$ in Definition 3, from which it is evident that Independence combines Within Source Independence and complete indifference among sources of risk. Therefore, like Independence, complete indifference among sources will also deliver an EU representation.

Under SREU, the preference among sources of risk can be captured by the relative curvature of their corresponding transformation functions. Such a characterization sets the stage for our derivation of its implications on the decision maker's attitude toward the processes of risk resolution. In particular, SREU delivers a *preference for decisiveness of preferred source* (PDPS) in the definition below and is characterized in following proposition.

Definition 4. *We say the decision-maker exhibits a preference for decisiveness of source s over source t if*

$$p_a a \oplus_s (1 - p_a) \left[\frac{p_b}{1 - p_a} b \oplus_t \frac{p_c}{1 - p_a} c \right] \succsim (1 - p_c) \left[\frac{p_a}{1 - p_c} a \oplus_s \frac{p_b}{1 - p_c} b \right] \oplus_t p_c c$$

for any $a, b, c \in \mathcal{M}$ such that $a \succsim b \succsim c$ or $c \succsim b \succsim a$, and for any $p_a, p_b, p_c \in (0, 1)$ with $p_a + p_b + p_c = 1$.

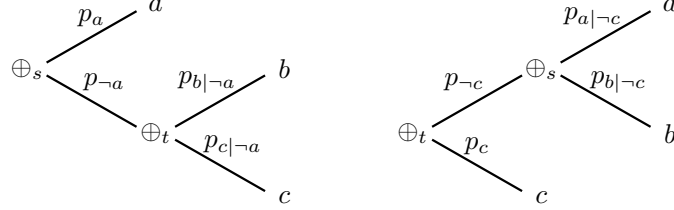


Figure 3: Preference for decisiveness of preferred source

The choice behavior in Definition 4 is illustrated using Figure 3 where $p_{\neg i} = 1 - p_i$ and $p_{i|\neg j} = p_i / (1 - p_j)$ for $i, j \in \{a, b, c\}$. The key to the PDPS interpretation lies in the assumption that b is intermediate in preference between a and c . Under this assumption, the utility distance between a and c is larger, therefore, the source of risk that determines which outcome between a and c obtains serves the decisive role. Notice that in the source lottery on the left of Figure 3, the source of risk s determines which outcome between a and c obtains; while in the source lottery on the right, the source t plays this role. Moreover, the sources of risk that split a and b are s in both source lotteries; the sources of risk that split b and c are t in both source lotteries. In the above senses, the difference between these two source lotteries lies only in the different sources of risk that play the decisive role. Therefore, preferring the source lottery on the left carries the interpretation of PDPS.

Proposition 1. *Under SREU, for any two sources of risk $s, t \in S$, the following statements are equivalent:*

- A. *The source s is weakly preferred to the source t .*

B. The decision-maker exhibits a preference for decisiveness of source s over source t .

C. The composite function $T_t \circ T_s^{-1}$ is weakly concave.

In Proposition 1, the equivalence between statements A and C follows from observing that the source s is preferred to the source t if and only if

$$T_t \circ T_s^{-1}(\alpha T_s \circ U(a) + (1 - \alpha) T_s \circ U(b)) \geq \alpha T_t \circ U(a) + (1 - \alpha) T_t \circ U(b)$$

for any $a, b \in \mathcal{M}$ and $\alpha \in [0, 1]$. That Statement B implies Statement A follows directly from observing that the former reduces to the latter in the limiting case of $p_b = 0$. To complete the proof of this proposition, we demonstrate, in an appendix, that Statement C implies Statement B.

The intuition of PDPS also manifests in the following proposition. The three preference relations all involve the preferred source of risk resolving more risk inherent in pairs of source lotteries. From the proposition, we also note that PDPS is different from the idea that the source lottery placing the preferred source in the first stage would be preferred.

Proposition 2. *Let $s, t \in S$ be two sources of risk. Suppose the source s is weakly preferred to the source t . Under SREU, for any $a, b, c \in \mathcal{M}$ such that $a \succsim b \succsim c$ or $c \succsim b \succsim a$, we have*

$$p_a a \oplus_s (1 - p_a) \left[\frac{p_b}{1 - p_a} b \oplus_t \frac{p_c}{1 - p_a} c \right] \succsim p_b b \oplus_s (1 - p_b) \left[\frac{p_a}{1 - p_b} a \oplus_t \frac{p_c}{1 - p_b} c \right]$$

for any $p_a, p_b, p_c \in (0, 1)$ with $p_a + p_b + p_c = 1$ and $p_a > p_b$;

$$p_b b \oplus_t (1 - p_b) \left[\frac{p_a}{1 - p_b} a \oplus_s \frac{p_c}{1 - p_b} c \right] \succsim p_a a \oplus_t (1 - p_a) \left[\frac{p_b}{1 - p_a} b \oplus_s \frac{p_c}{1 - p_a} c \right]$$

for any $p_a, p_b, p_c \in (0, 1)$ with $p_a + p_b + p_c = 1$ and $p_a > p_b$; and

$$p_b b \oplus_t (1 - p_b) \left[\frac{p_a}{1 - p_b} a \oplus_s \frac{p_c}{1 - p_b} c \right] \succsim p_b b \oplus_s (1 - p_b) \left[\frac{p_a}{1 - p_b} a \oplus_t \frac{p_c}{1 - p_b} c \right]$$

for any $p_a, p_b, p_c \in (0, 1)$ with $p_a + p_b + p_c = 1$ and $\min\{p_a, p_c\} > p_b$.

Relative to their respective right-hand side source lotteries, all three source lotteries on the left place the preferred source s in determining which outcome between a and c obtains, with the largest utility discrepancy among the three outcomes. Following the intuition of PDPS, they should be preferred. However, there are boundary constraints for PDPS to hold. In the first two pairs of source lotteries, the ones on the right place the preferred source s in determining which outcome between b and c obtains. Though the utility difference b and c is smaller, for PDPS to manifest, the boundary condition $p_a > p_b$ is needed to ensure that the realization between a and c is more decisive. In the source lottery on the left in the third preference relation, the source t determines which between b and a obtains as well as which between b and c obtains. This is replaced by the preferred source s in the source lottery on the right. Therefore, the stronger boundary condition $\min\{p_a, p_c\} > p_b$ is required for the intuition of PDPS to apply.

Indifference between two sources of risk s and t is equivalent to an across-source associativity property between them: for any $a, b, c \in \mathcal{M}$, and for any

$p_a, p_b, p_c \in (0, 1)$ with $p_a + p_b + p_c = 1$:

$$p_a a \oplus_s (1 - p_a) \left[\frac{p_b}{1 - p_a} b \oplus_t \frac{p_c}{1 - p_a} c \right] \sim (1 - p_c) \left[\frac{p_a}{1 - p_c} a \oplus_s \frac{p_b}{1 - p_c} b \right] \oplus_t p_c c.$$

Therefore, across-source associativity between any pair of sources is equivalent to complete indifference among sources, offering another way to pinpoint how we have relaxed the EU assumptions. Mixture Reduction on a standard mixture set is implied by an associativity property (defined by replacing $\oplus_s$ and $\oplus_t$ with $\oplus$ in the above equation) and a reducibility property ($\lambda a \oplus (1 - \lambda)a = a$ for $\lambda \in [0, 1]$) (see Mongin, 2001). In a different vein, Rommeswinkel (2023) generalizes the standard mixture set by dropping the reducibility property to capture a value of mixing per se.

Across-source associativity between two sources of risk s and t provides a specific perspective to see how the risk resolution process is irrelevant to a decision maker. The two source lotteries on the two sides of the indifference relation both reduce to a lottery delivering outcome $x \in \{a, b, c\}$ with probability p_x . Their difference lies only in the risk resolution process. In this sense, we can regard the equivalence between indifference between two sources s and t and the across-source associativity between them as a case of the more general equivalence between source preference and process preference. Next, we provide an account of this more general equivalence by bringing in a state-space foundation.

Let the state space be $S = \{s_1, s_2, s_3, s_4\}$. On this state space, $E_1 = \{s_1, s_2\}$ and $E_2 = \{s_3, s_4\}$ are events from a source of risk s , while $F_1 = \{s_1, s_3\}$ and

$F_2 = \{s_2, s_4\}$ are events from another source of risk t . We have probabilities $p(s_1) = r$, $p(s_2) = p - r$, $p(s_3) = q - r$, and $p(s_4) = 1 - p - q + r$. Consider the following two source lotteries

$$P = p\left[\frac{r}{p}x_1 \oplus_t \frac{p-r}{p}x_2\right] \oplus_s (1-p)\left[\frac{q-r}{1-p}x_3 \oplus_t \frac{1-p-q+r}{1-p}x_4\right]$$

and

$$Q = q\left[\frac{r}{q}x_1 \oplus_s \frac{q-r}{q}x_3\right] \oplus_t (1-q)\left[\frac{p-r}{1-q}x_2 \oplus_s \frac{1-p-q+r}{1-p}x_4\right].$$

Both source lotteries correspond to the same Savagian act that delivers x_i with $i \in \{1, 2, 3, 4\}$ on state s_i and the same lottery. They differ in the risk resolution process by switching the order of the sources of risk. We have the following proposition.

Proposition 3. *Under SREU, the following two statements are equivalent:*

1. *For any $x_1, x_2, x_3, x_4 \in \mathcal{M}$, any $p, q \in (0, 1)$, and any $r \in [0, \min\{p, q\}]$, $P \sim Q$.*
2. *The source s is indifferent to the source t .*

Notice that across-source associativity between two sources of risk s and t is a special case of Statement 1 by holding $r = 0$. Though the proposition only involves four outcomes, it essentially proves the equivalence between source preference and process preference under SREU, i.e., for non-trivial process preference to vanish, source preference must also disappear, and vice versa.

5 Experiment and Application

5.1 Testing PDPS

Subjects are asked to complete two types of tasks (in this order): i) a PDPS task and ii) source preference tasks. Both involve flipping face-down cards (red or green) and two sources of risk: i) flipping themselves, or ii) partner flipping. For a source preference task, there is one red card and one green card. Subjects choose whether to flip a card themselves or have their partner flip a card. As shown in Figure 4, the payoff distribution is identical regardless of who flips the card. However, subjects receive the higher payoff if they flip the red card themselves, and the lower payoff if their partner flips it. Subjects complete two source preference tasks, with payoffs differing between tasks. We say a subject exhibits source preference if they consistently choose the same source.

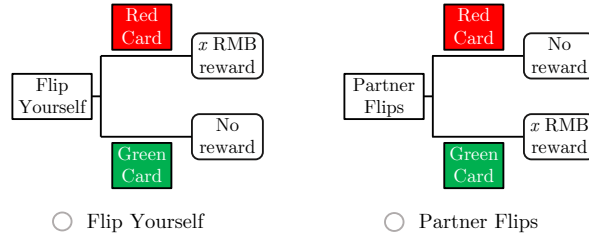


Figure 4: Source preference tasks, $x \in \{10, 20\}$

For a PDPS task, there is one red card and two green cards. Subjects choose whether to flip a card themselves first (and their partner second) or have their partner flip a card first (and themselves second). The payoff distribution is

identical regardless of the order in which cards are flipped. Subjects receive the higher payoff if they flip the red card themselves, and the lower payoff if their partners flip the red card. If the red card is never flipped, subjects receive an intermediate payoff. Subjects complete one PDPS task as shown in Figure 5. Conditional on having a preferred source, we say a choice is consistent with PDPS if the preferred source is chosen such that it is in the first stage of the source lottery.

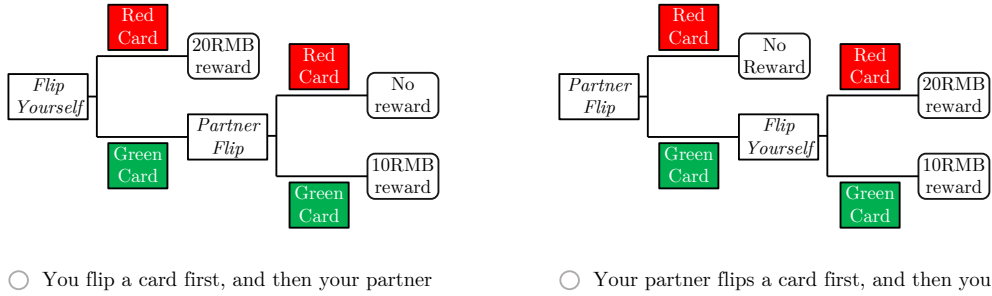


Figure 5: Preference for decisiveness of preferred source task

The experiment was coded in o-Tree (Chen et al., 2016) and conducted in the laboratory at the China Center for Behavioral Economics and Finance at the Southwestern University of Finance and Economics with a total sample size of 190 (11 sessions with 16 subjects and 1 session with 14 subjects).⁹ The average payment per subject (including show-up fee of RMB10) was approximately RMB40 for a 15 minute session (equivalent to approximately \$22.50 per hour). All of the tasks are performed on a computer. The PDPS task and a randomly chosen source preference task (by computer) determine the

⁹We also collected data on basic demographics, college major, and frequency of past participation in experiments without imposing any exclusion criteria. Our experiment was post-registered on the AEA Registry (AEARCTR-0014036).

Preferred source			
	Flip yourself first	Partner flip first	Total
Flip yourself	68 [33]	32 [21]	100 [54]
Partner flip	19 [6]	13 [9]	32 [15]
Neither	33 [14]	25 [12]	58 [26]
Total	120 [53]	70 [42]	190 [95]

Table 1: The table shows the cross-tabulation of the number of subjects by their preferred source and whether they choose to flip first/partner flip first. The table also shows the number of choices for those who did not exhibit source preference. The numbers in brackets are the number of subjects after screening.

subject’s total payment. The results are summarized in Table 1.

To test whether PDPS holds, we need to focus on subjects who exhibit source preference. Of the 190 subjects, 132 ($\approx 70\%$) display such a preference: 100 prefer to flip the cards themselves, and 32 prefer their partner to flip. We use the most conservative null hypothesis, which assumes that these subjects choose randomly between flipping the cards themselves first and having their partner flip first, implying equal proportions of subjects choosing according to PDPS and those who do not. The one-sided alternative hypothesis is that the proportion of subjects who satisfy PDPS is greater than those who do not, i.e. that more than 50% of the subjects satisfy PDPS.¹⁰

Of the 100 who exhibit source preference for flipping themselves, 68 preferred flipping themselves first over their partner and among the 32 who exhibit source preference for their partner flipping, 13 preferred their partner first over

¹⁰This is the most conservative null hypothesis because any percentage lower than 50% would consistently yield a lower p-value.

themselves. This results in 81 out of 132 ($\approx 61\%$) behaving in accordance with PDPS. Using a standard one-sided binomial test, we reject the null hypothesis ($p = 0.005657 < 0.001$) in favor of PDPS at any reasonable significance level.

To ensure the robustness of our main finding regarding PDPS, we incorporate an understanding question about the PDPS task. We ask subjects if the probability distribution over outcomes of receiving RMB0, RMB10, and RMB20 is: a) $(1/3, 1/3, 1/3)$, b) $(1/2, 1/4, 1/4)$, c) $(1/4, 1/2, 1/4)$, d) $(1/2, 1/4, 1/4)$, or e) *other*. After this screening, 33 out of 54 subjects who prefer flipping themselves and 9 out of 15 who prefer their partner to flip choose in line with PDPS, totaling 42 out of 70 subjects (60%). The result of PDPS remains robust after screening as we reject the null ($p = 0.045593$) in favor of PDPS at 5% significance level. A further robustness check is related to those who do not exhibit source preference. Such subjects should choose randomly (i.e. 50%) between flipping themselves first or their partner flipping first. We do not reject this hypothesis at any reasonable significance level ($p = 0.179072/0.422509$, before/after screening).

Consistent with the literature in source preference and “*illusion/locus of control*” (e.g. Li, 2011), we might expect PDPS to be more evident when the source is subjects flipping cards themselves. To investigate this, we conduct the Conlisk test (Conlisk, 1989) for PDPS which allows us to test for PDPS behavior with PDPS being more frequent with subjects flipping cards themselves than their partner flipping cards. We reject the null hypothesis ($p \ll 0.01$) in favor of PDPS being more frequently observed for subjects who prefer to flip the cards themselves at any reasonable significance level. Repeating the

one-sided binomial tests as above, conditional on each source, is also in line with the Conlisk test.

We provide novel evidence in favor of behavior that is consistent with PDPS when decision-makers exhibit source preference, in which we view self-flipping and partner-flipping of cards as different sources of risk. From this perspective, our paper is related to the strand of literature focusing on “locus of control” (Charness and Gneezy, 2010; Li, 2011). For instance, Charness and Gneezy (2010) find that the majority of subjects (68%) exhibit a preference for rolling a die themselves over the experimenter. In explaining our experimental result, we observe that subjects tend to prefer the self-flipping source over partner-flipping source and also exhibit PDPS in deciding the sequence of card flips.

5.2 Home Bias

In international finance, explaining the persistence of home bias remains an active area of interest (see, e.g., Guidolin and Liu, 2016; Dimmock et al., 2016; Berger and Eeckhoudt, 2006). One direction of research, following Coval and Moskowitz (1999) and Huberman (2001), seeks a preference-based explanation (see, e.g., Boyle et al., 2012; Solnik and Zuo, 2012; Asano and Osaki, 2020; Cappelli et al., 2020). We contribute to this direction of research by modeling the investor as perceiving her portfolio in terms of compound risks comprising domestic and foreign markets as distinct sources of risk.

Suppose that an investor faces a portfolio choice problem in allocating her total assets between the domestic stock market (D) and the foreign stock market (F). These are two sources of risk. For simplicity, suppose that when

the investor allocates all her assets to one of the two markets, it yields either \$1 or \$0 with the same probabilities of 50%. The two markets are positively, but not perfectly, correlated, i.e., $0 \leq \alpha < 1/2$ where α is the probability that both markets yield \$1.

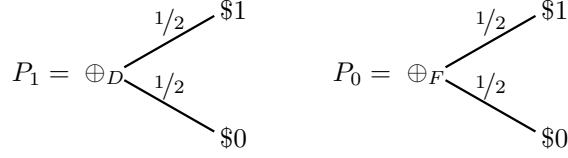


Figure 6: Domestic and foreign investments

Denote a portfolio with θ proportion of domestic stocks by P_θ . If an SREU investor is indifferent between the two sources of risk, RCLA applies to the compounding of risks when diversifying between domestic and foreign stocks. Consequently, the utility of the portfolio P_θ is given by

$$U(P_\theta) = \alpha(u(1) + u(0)) + (1/2 - \alpha)(u(\theta) + u(1 - \theta))$$

in which $u : [0, 1] \rightarrow \mathbb{R}$ is a utility function of investment returns. In this case, the transformation functions for sources D and F can both be fixed as the identity function $T_D(x) = T_F(x) = x$. Suppose u is strictly increasing and strictly concave, capturing the investor's monotonicity and risk aversion towards investment returns, then her utility is maximized when she perfectly diversifies her portfolio, i.e., $\theta^* = \frac{1}{2}$. In other words, the investor does not exhibit home bias when indifferent between sources D and F .

To apply SREU when the investor exhibits a preference between the sources D and F , we need to be specific about the portfolio's risk resolution process.

There are two possible specifications, as depicted in Figure 7. We can show that the home bias result of the following Proposition 4 is robust to either specification. Therefore, without loss of generality, we continue our analysis with the one on the right, positioning the source D at the first stage.

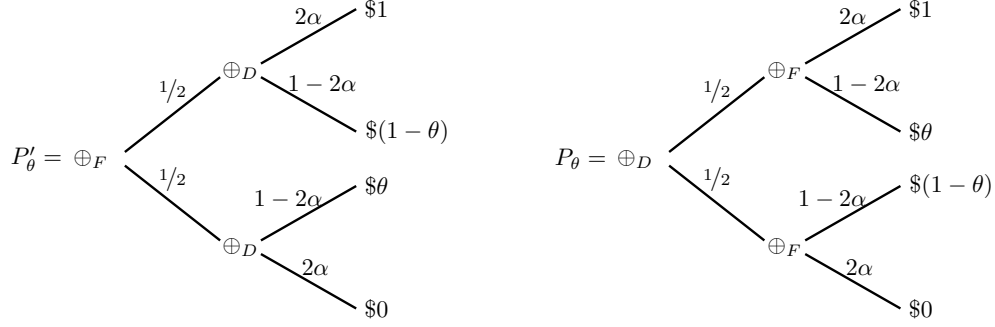


Figure 7: Diversified portfolio

The utility $U(P_\theta)$ is evaluated as if the investor first derives the certainty equivalents for the two lotteries at the second stage, both of which arise from source F , with one delivering either \$1 or θ and the other delivering either $\$(1 - \theta)$ or \$0. Their certainty equivalents c_θ and d_θ are given by $u(c_\theta) = T_F^{-1}(2\alpha T_F \circ u(1) + (1 - 2\alpha)T_F \circ u(\theta))$ and $u(d_\theta) = T_F^{-1}((1 - 2\alpha)T_F \circ u(1 - \theta) + 2\alpha T_F \circ u(0))$, respectively. The investor then evaluates the diversified portfolio as a 50-50 lottery from source F delivering c_θ and d_θ as outcomes:

$$U(P_\theta) = 0.5u(c_\theta) + 0.5u(d_\theta).$$

Under the assumption of a strict preference for the domestic stock market, we have a strictly concave T_F (with the transformation function for source D being the identity function). Assuming that u and T_F are differentiable, we

arrive at the following proposition whose proof is provided in the appendix.

Proposition 4. *Suppose a risk-averse SREU investor strictly prefers the source D over F , then she exhibits home bias, i.e., $\theta^* \in (0.5, 1)$ where θ^* maximizes $U(P_\theta)$.*

It is worth pointing out that the distinct specifications of the portfolio allow us to model how the investor might mentally represent or perceive the portfolio. This factor of choice perception in decision making has received increasing attention in the recent literature (Epstein and Halevy, 2019; Esponda and Vespa, 2023; Chew et al., 2023).

6 Conclusion

To accommodate compounding risks from multiple sources of risk, we extend Herstein and Milnor’s (1953) mixture set axiomatization of EU to source mixture sets to characterize SREU by relaxing mixture reduction and independence with their source-based counterparts. This approach allows for non-RCLA behavior across different sources of risk due to the failure of across-source associativity. We present experimental evidence supporting PDPS and demonstrate a new explanation for home bias based on source preference.

One direction of follow-up theory research is to generalize SREU by adopting non-EU preferences for risks arising from the same source. In this regard, weighted utility (Chew, 1983) offers a candidate since it exhibits RCLA.¹¹ An-

¹¹This follows from observing that weighted utility can be written as the ratio of two mixture-linear functions.

other direction is to explore the implications of further generalizing our source mixture set in the direction outlined by Rommeswinkel (2023). This line of investigation offers a potential foundation for modeling the idea of recreational utility, which may arise from a source-based process preference together with an intrinsic value of mixing.

In terms of applied research, we may add SREU to the toolkit to address market phenomena involving the compounding of risks from multiple sources potentially arising from heterogeneity in market participants. These include non-participation and under-diversification in financial markets besides home bias and the increasingly diverse and uncertain employment choices in today's increasingly volatile labor markets. Additionally, many people may view the uncertainties brought by AI's growing role in society with caution, such as patients holding idiosyncratic attitudes toward human-only versus AI-assisted healthcare delivery.

Appendix: Proof

Proof of Theorem 1: For each $s \in S$, $(\mathcal{M}, \oplus_s)$ is a mixture set and the preference ordering $\succsim$ satisfies Herstein and Milnor's (1953) three axioms on $(\mathcal{M}, \oplus_s)$. This yields a von Neumann-Morgenstern utility function $U_s : \mathcal{M} \rightarrow \mathbb{R}$ representing $\succsim$ such that

$$U_s(\alpha a \oplus_s (1 - \alpha)b) = \alpha U_s(a) + (1 - \alpha)U_s(b)$$

for all $a, b \in \mathcal{M}$ and all $0 \leq \alpha \leq 1$. Fix a source $s^* \in S$ and choose U_{s^*} such that there exists $a^* \in \mathcal{M}$ with $U_{s^*}(a^*) = 0$. Let $U = U_{s^*}$ be the utility function in the SREU representation and the transformation function for the source s^* be the identity function $T_{s^*}(x) = x$. For each $s \in S$ with $s \neq s^*$, choose U_s such that $U_s(a^*) = 0$ and define T_s such that $T_s \circ U(a) = U_s(a)$ for each $a \in \mathcal{M}$. Then, T_s is uniquely defined on the interval $I = U(\mathcal{M})$ with $T_s(0) = 0$, and strictly increasing on I since each U_s represents the same preference ordering as U . Furthermore, T_s is continuous on I . To see why, consider $a, b, c, d \in \mathcal{M}$ such that $b \sim \alpha^* a \oplus_s (1 - \alpha^*)d$ and $c \sim \alpha^* a \oplus_{s^*} (1 - \alpha^*)d$ for some $\alpha^* \in [0, 1]$. Let $\{\alpha_n\}$ be any sequence of real numbers in $[0, 1]$ converging to α^* . By Axiom 2, $\{U_s(\alpha_n a \oplus_s (1 - \alpha_n)d)\}$ converges to $U_s(b)$ and $\{U(\alpha_n a \oplus_{s^*} (1 - \alpha_n)d)\}$ converges to $U(c)$. Since a, b, d and $\{\alpha_n\}$ can be chosen arbitrarily, the image of the limit of any converging sequence under T_s is the limit of their images. Finally, notice that $\{T_s\}_{s \in S}$ intersect with each other at 0.

The converse is straightforward. □

The proof of Proposition 1 makes use of the following lemma which is

stated without proof.

Lemma 1. *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ denote a strictly increasing, weakly concave function with $g(0) = 0$. If $x_1 x_2 \leq 0$, then $g(x_1 + x_2) \geq g(x_1) + g(x_2)$.*

Proof of Proposition 1: (C $\implies$ B) To simplify exposition, we only prove that the preference ordering holds when $U(b) = 0$. Apply the main Theorem to Statement B, yielding the following inequality to be shown based on Statement C.

$$T_s^{-1} \left(p_a T_s \circ U(a) + (1 - p_a) T_s \circ T_t^{-1} \left(\frac{p_c}{1 - p_a} T_t \circ U(c) \right) \right) \geq \\ T_t^{-1} \left(p_c T_t \circ U(c) + (1 - p_c) T_t \circ T_s^{-1} \left(\frac{p_a}{1 - p_c} T_s \circ U(a) \right) \right)$$

Letting x , and y , and $f(\cdot)$ denote $T_s \circ U(a)$, $T_t \circ U(c)$, and $T_t \circ T_s^{-1}(\cdot)$ respectively, the above inequality simplifies to:

$$f \left(\underbrace{(1 - p_a) f^{-1} \left(\frac{p_c}{1 - p_a} y \right)}_Y + \underbrace{p_a x}_X \right) \geq \underbrace{p_c y}_{Y'} + \underbrace{(1 - p_c) f \left(\frac{p_a}{1 - p_c} x \right)}_{X'} \quad (1)$$

By concavity of f , we have $f(Y) \geq (1 - p_a) f \circ f^{-1}(\frac{p_c}{1 - p_a} y) = Y'$ and $f(X) \geq X'$. Adding the two inequalities yields $f(Y) + f(X) \geq Y' + X'$, so that $f(Y + X) \geq Y' + X'$ by Lemma 1, as required to complete the proof. $\square$

The lemma below, without a proof, is used in the proof of Proposition 4.

Lemma 2. *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be strictly increasing and strictly convex. If $x_1 + y_1 = x_2 + y_2$ and $|x_1 - y_1| > |x_2 - y_2|$, then $g(x_1) + g(y_1) > g(x_2) + g(y_2)$.*

Proof of Proposition 4: Observe that c_θ and d_θ , defined in Section 5.2, satisfy $\theta < c_\theta < 1$ and $0 < d_\theta < 1 - \theta$ for each θ . Notice that $T_F \circ u(c_\theta) + T_F \circ u(d_\theta) = T_F \circ u(c_{1-\theta}) + T_F \circ u(d_{1-\theta})$ for each $\theta \in [0, 1]$. Let $A = 2\alpha[T_F \circ u(1) - T_F \circ u(0)]$ and $B = (1 - 2\alpha)[T_F \circ u(\theta) - T_F \circ u(1 - \theta)]$. Then, $T_F \circ u(c_\theta) - T_F \circ u(d_\theta) = A + B$ and $T_F \circ u(c_{1-\theta}) - T_F \circ u(d_{1-\theta}) = A - B$. A is clearly positive and for $0.5 < \theta \leq 1$, B is also positive. This means that $|A + B| > |A - B|$. Applying Lemma 2 for the strictly convex inverse transformation function T_F^{-1} , we have $U(P_\theta) > U(P_{1-\theta})$ for each $0.5 < \theta \leq 1$. Thus, any utility maximizing θ^* satisfies $0.5 \leq \theta^* \leq 1$.

The derivative for U_θ is given by

$$U'(\theta) = \frac{(1 - 2\alpha)}{2} \frac{T'_F \circ u(\theta)u'(\theta)}{T'_F \circ u(c_\theta)} - \frac{(1 - 2\alpha)}{2} \frac{T'_F \circ u(1 - \theta)u'(1 - \theta)}{T'_F \circ u(d_\theta)}.$$

Due to the strict concavity of T_F and u , the following inequalities hold:

$$U'(0.5) = \frac{(1 - 2\alpha)T'_F \circ u(0.5)u'(0.5)}{2} \left(\frac{1}{T'_F \circ u(c_{0.5})} - \frac{1}{T'_F \circ u(d_{0.5})} \right) > 0;$$

$$U'(1) = \frac{(1 - 2\alpha)}{2} (u'(1) - u'(0)) < 0.$$

Thus, θ^* cannot be 0.5 nor 1. This completes the proof. □

Appendix: Experiment Instructions

The original instructions and interfaces are in Chinese. In this appendix, we attach some translated screenshots of our programme.

Experiment Instruction

Welcome to the experiment. You will receive incomes from two games plus a 10 RMB show-up fee.

Now, we introduce the rules of the first game.

In this game, there are three face-down cards. Only one card (randomly positioned) is red and the other two are green.

All participants are assigned to groups of two. You and your teammate flip the three cards in sequence. The game ends when the red card is flipped or only one card remains unflipped, and you will be paid accordingly.

We will first play two practice rounds.

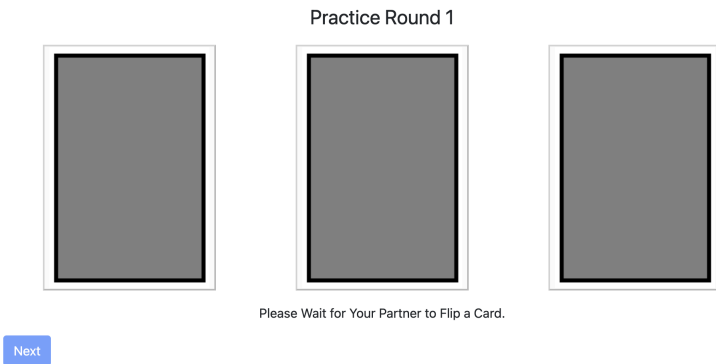
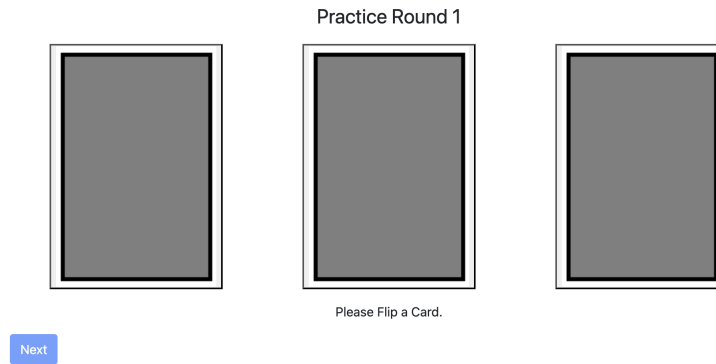
In the first practice round, you flip a card first, then your partner flips. In the second round, your partner flips first, then you flip.

When the two practice rounds are over, participants will be reassigned into new groups for the formal game.

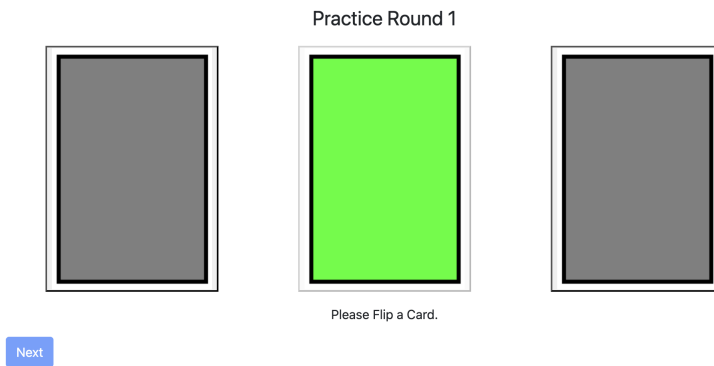
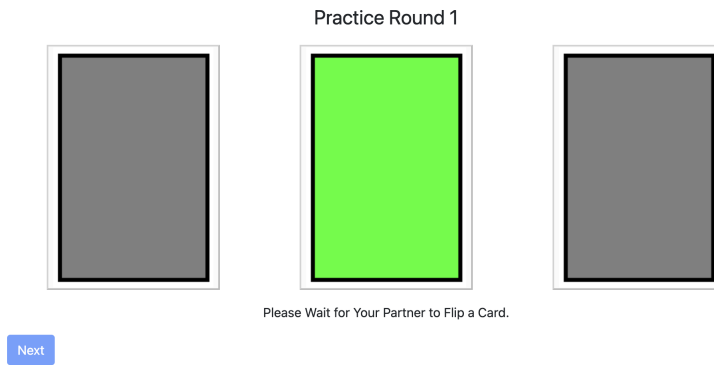
Next

Note that this is the first instruction page for half of participants in a session. For the other half, the order to flip cards is switched in the two practice rounds.

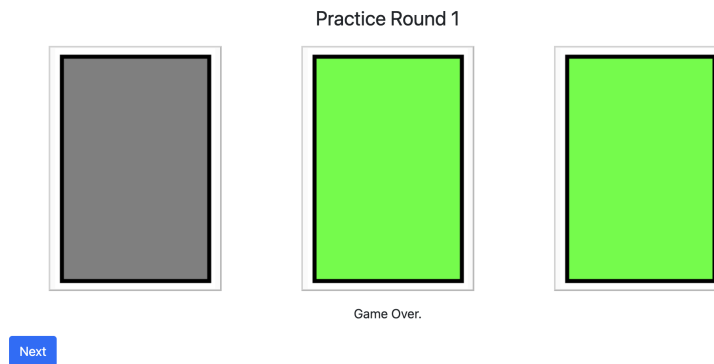
We illustrate how participants play the game, i.e., sequentially flipping cards on computers, using an example of the first practice round.



When both group members enter the first practice round, they are presented three face-down cards. One member is called to flip a card and the other member is told to wait for her partner. The active member can click on a card to flip it. Its color is revealed to both members.



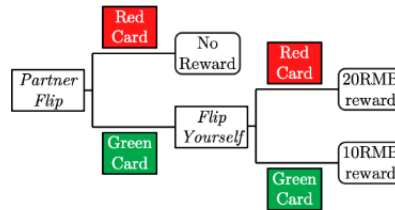
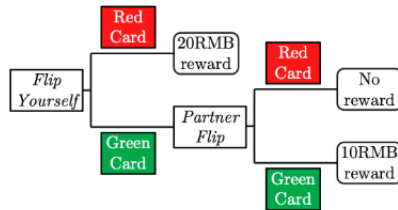
When the red card is flipped or both green cards are flipped, both group members are notified that the game ends and the 'Next' button is enabled.



After the two practice rounds, the subjects are asked to select who will flip a card first and to assess the probabilities of getting different rewards under different scenarios. The formal game starts after this page.

Specify the Rule of Your Game

As illustrated in the following figures, if you flip the red card, you will be rewarded 20 RMB; if your partner flips the red card, there is no reward; and if the red card is not flipped, you will be rewarded 10 RMB. Please select who will flip first.



☐ You flip a card first, and then your partner

☐ Your partner flips a card first, and then you

If you choose to flip a card first, what are the probabilities of getting 20 RMB, 10 RMB, or nothing?

- ☐ A. 33.3...%, 33.3...%, 33.3...%
- ☐ B. 50%, 25%, 25%
- ☐ C. 25%, 50%, 25%
- ☐ D. 25%, 25%, 50%
- ☐ E. Other probabilities

If you choose to let your partner flip a card first, what are the probabilities of getting 20 RMB, 10 RMB, or nothing?

- ☐ A. 33.3...%, 33.3...%, 33.3...%
- ☐ B. 50%, 25%, 25%
- ☐ C. 25%, 50%, 25%
- ☐ D. 25%, 25%, 50%
- ☐ E. Other probabilities

Next

After implementing the first game, subjects read the following instruction for the second game and choose who will flip a card. The game is also played on computers.

Experiment Instruction

Now, we introduce the rules of the second game.

In this game, you will see two face-down cards. One of them is red, and the other one is green.

All participants are regrouped into pairs. You can choose who will flip a card, and you will be rewarded accordingly.

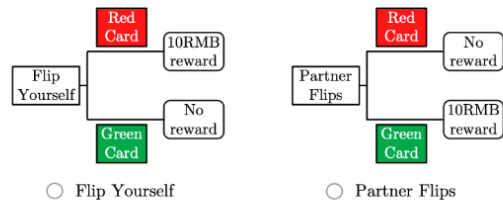
You will specify the rules of two games. A random one will be chosen to determine your income in the second game.

Next

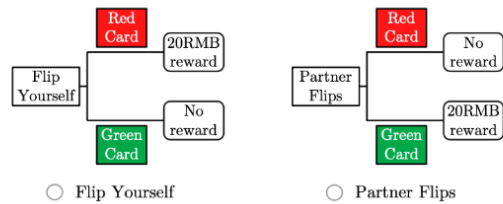
Specify the Rule of Your Game

The program will randomly implement one of the following two games.

If you flip a red card, you will be rewarded 10 RMB. If your partner flips a green card, you will be rewarded 10 RMB.



If you flip a red card, you will be rewarded 20 RMB. If your partner flips a green card, you will be rewarded 20 RMB.



Next

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